

In the third generation of quantum dot televisions, some particles (metal alloy core and metal alloy shell) have been changed. This technology has enabled far better visibility with peak brightness and colour accuracy. Further, it claims to give viewers a good visual experience.

Construction of an image

There are three basic colours—namely, red, green and blue—that are used to create a colour image. Consider quantum dot LEDs in which a thin layer of quantum dots is used in front of the backlight. In LCD televisions a liquid crystal layer is used to generate different colour spectrums. In QLED a layer of quantum dots is used. As shown in Fig. 2, six layers are used to construct a colour image. Here, functions of all the layers are given in brief.

The backlight unit illuminates the LED from the side or back of the display panel as QLEDs do not produce light by themselves. These need a special source of light to produce a visible image. OLEDs (organic LEDs) illuminate the LEDs by themselves and hence do not need any light source. Light produced by the backlight unit for QLEDs is passed to the reflector, which enhances the uniformity of the illuminance distribution on the target region of the image plane.

A polarising filter is used to change the settings of the image passed by the reflector so as to suppress the glare or blur the image. There are various types of polarisers used in display technology. The importance of a thin-film-transistor (TFT) array is to match the size with the quantum dot in the colour filter (in QLED) and carry out colour reproduction. It also handles how the pixels are displayed on the screen. In QLED, quantum dot colour filters coupled with a blue LED can produce billions of colours with peak brightness and better clarity.

There are two main key advantages of quantum dot materials. First,

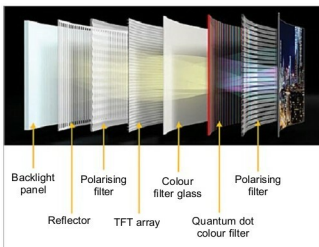


Fig. 2: Image construction (Source: <https://lytimg.com>)

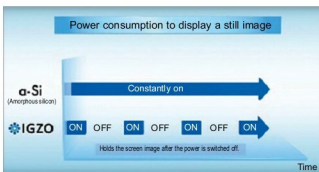


Fig. 3: Low power consumption IGZO versus a-Si (Source: <https://android.gadgethacks.com>)

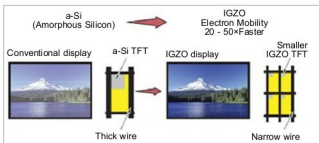


Fig. 4: High-resolution IGZO versus a-Si (Source: sharpsma.com)

these resist moisture better and thus have a longer life period. Second, because of their higher resistance to moisture, these do not require an expensive vacuum evaporation process. Last, a polarising filter gives a final touch to the display to produce a crystal-clear image on the screen.

IGZO technology

IGZO is composite of indium (Id), gallium (Ga), zinc (Z) and oxygen (O). An IGZO-TFT is artificially produced and used in flat panel displays, which are made of transparent crystalline oxide semiconductors. This newly-invented IGZO-TFT enables LCD and OLED displays to achieve new

levels of performance. Because of its attractive features, it is being used in ASUS notebooks, ultrabooks by Fujitsu and Razor Blade gaming laptops. Sharp mass produces it successfully. Low power consumption with high performance and high resolution are the key features of this technology.

Low power consumption with high performance

A comparison between the conventional display method and new display methods is shown in Fig. 3. The former consumes more power because all pixels are driven continuously. IGZO consumes less power for back-lighting or illumination in LED panel as compared to a-Si.

There are a number of TFTs arranged in a particular structure which will switch on or off depending on the contents being displayed on the screen. If the static image or video content is being displayed using IGZO that operates in pause mode (as it is in this case), there will be less leakage current and no flicker effect, which results in low power consumption. Thus, in IGZO-TFT technology, pixels maintain their charge throughout the cycle, which results in low power consumption and flicker-less high performance.

High resolution

In IGZO-TFT, as shown in Fig. 4, transistor size is small as compared to a-Si-TFT. Because of that, more numbers of pixels per area can be accommodated as compared to a-Si, which enables high resolution. On top of this, IGZO has a much lower leakage current and 20 to 50 times higher electron mobility as compared to a-Si, which results in fast response time or less processing delay.

To summarise, new display techniques using quantum dots and IGZO have changed the entire world of displays, especially in television domain. Their extraordinary features enable these to consume low power, perform better, give HD colour resolution, and televisions to be made thinner, lighter and more energy efficient. **EFY**